Challenge 1

Build a ETL

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Constitution

Mision

A company need to applies a 10% discount to the first 500 online orders placed through the platform. Only the first order per customer per day counts toward the threshold.

Tech Stack

- Data extraction via kaggle hub

- Jupyter Notebook built with Python (pandas, NumPy, SQLAlchemy)

- PostgreSQL database hosted on Neon serverless, connected via psycopg2

Roadmap

- Extract the dataset from the Kaggle platform

- Build a Jupyter notebook ETL pipeline (Extract → Transform → Load)

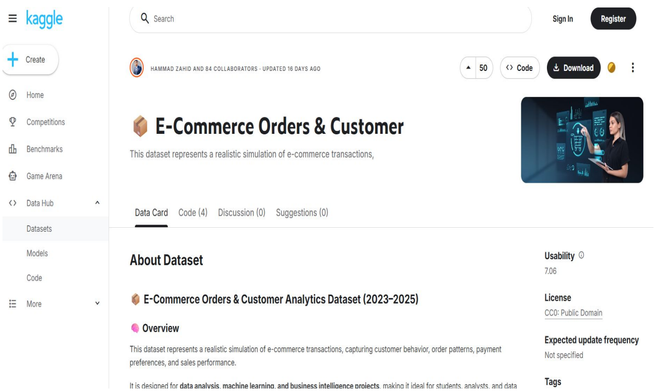
- Apply discount business logic and load data into a Neon PostgreSQL database using a medallion architecture (Bronze / Silver / Gold)

Feature Phase

Phase 1

Task 1

Download the e-commerce dataset from Kaggle using kaggle hub and cache it locally



Task 2

Rename columns, cast types, deduplicate by (date, customer), and apply the 10% discount logic for the first 500 online orders.

Feature Phase

Phase 1

Task 3

Write data through a medallion architecture:

- Bronze: raw deduplicated data (raw_ventas)

- Silver: enriched with discount columns (silver_ventas)

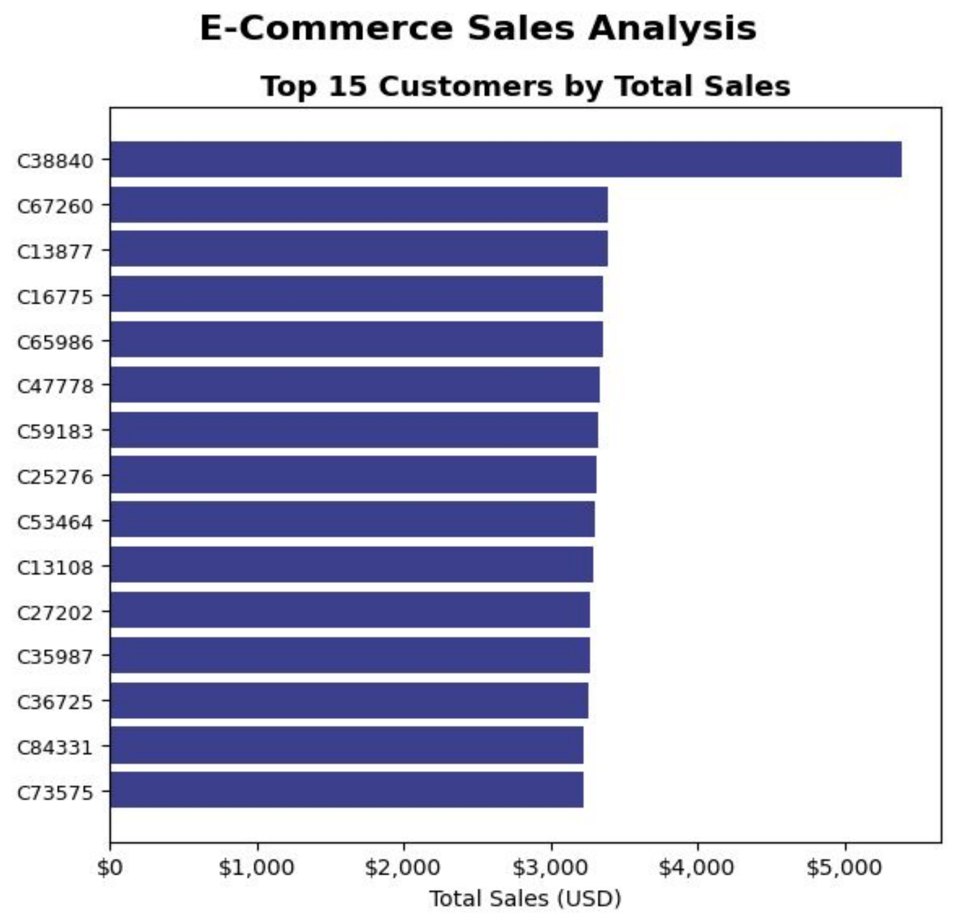
- Gold: star schema (dim_cliente, dim_produto, fact_ventas)

Task 4

Query the star schema on Neon for business metrics: sales by customer, by product, and by day.

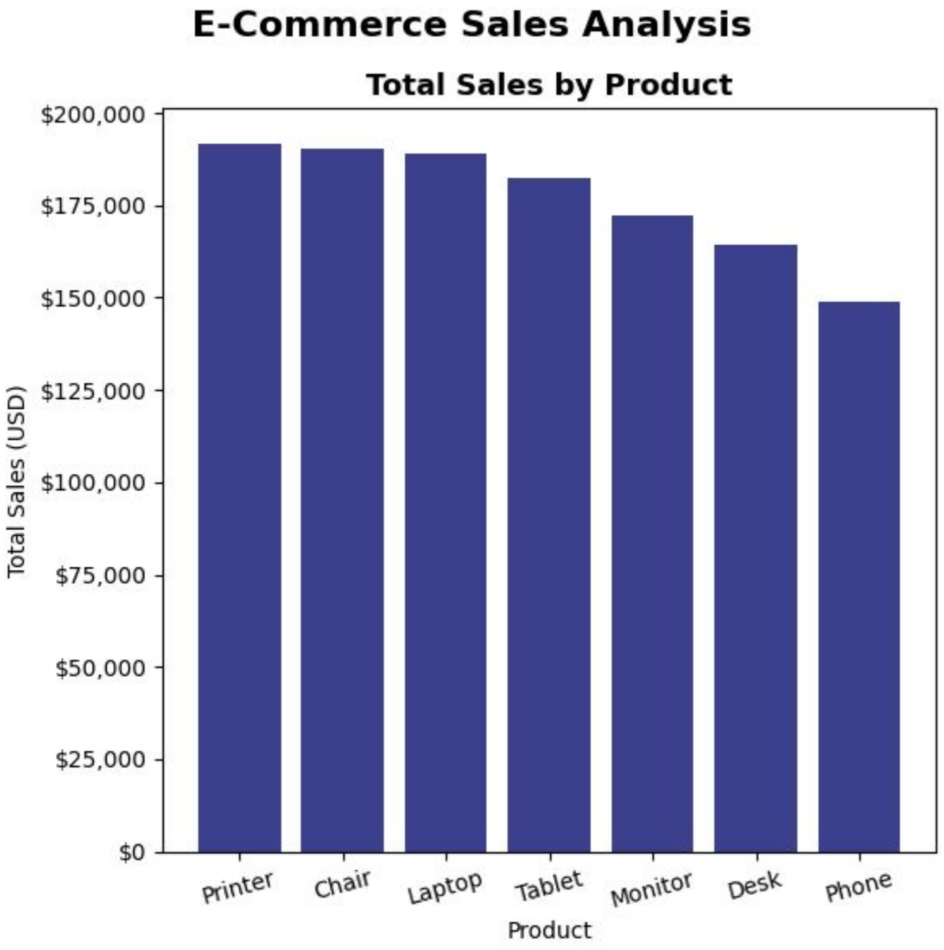
Total sales by customer

Ranks customers by total amount spent (final_total) across all their orders.



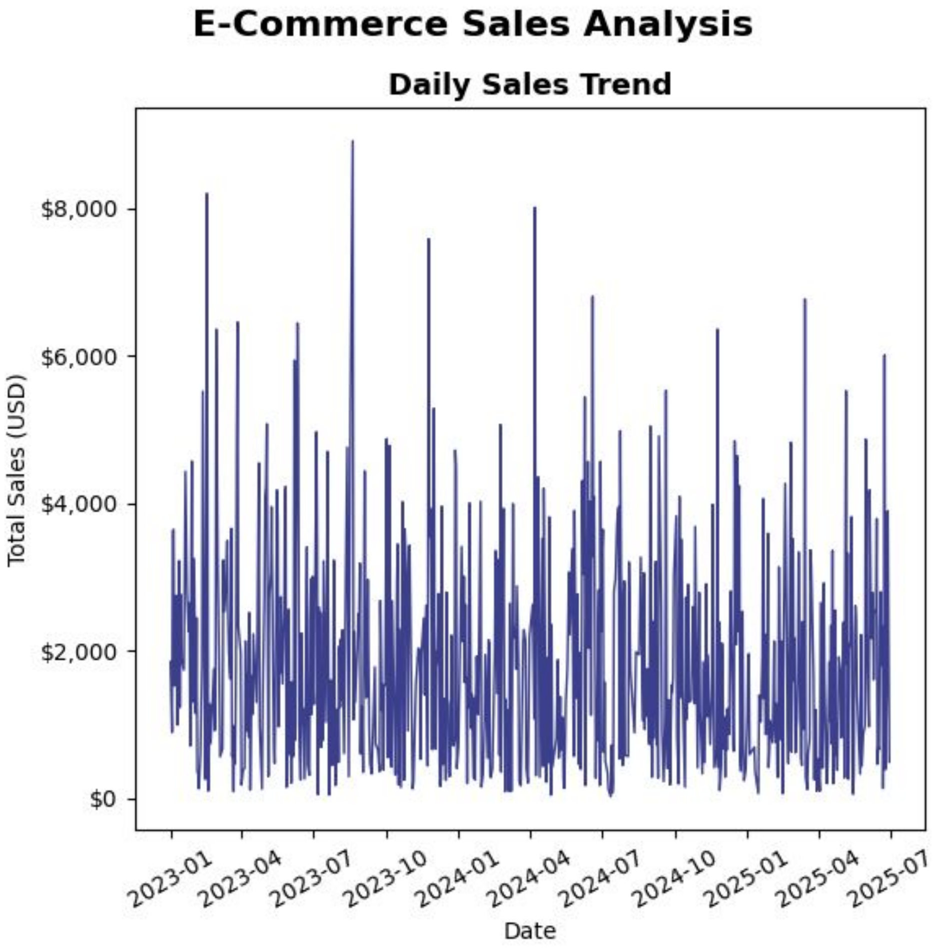
Total sales by product

Ranks products by total revenue generated



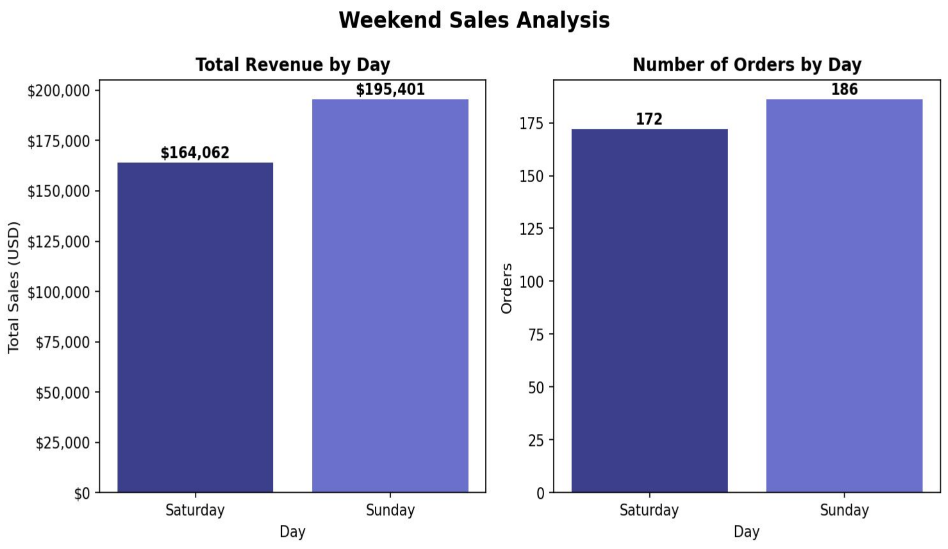
Sales by day

Daily revenue totals ordered chronologically — useful for trend and seasonality analysis.



Weekend Sales Analysis

Total revenue on weekends only, broken down by day name (Saturday / Sunday), joined through dim_fecha.



Thanks